

Supplementary Information for *CubeDynamics: An Inspectable Grammar for Environmental Data Analysis*

Ty Tuff

Environmental Data Science Innovation & Impact Lab
Cooperative Institute for Research in Environmental Sciences
University of Colorado Boulder, Boulder, Colorado, USA
[ORCID 0000-0001-5249-5197](https://orcid.org/0000-0001-5249-5197)

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Abstract

This Supplementary Information documents how CUBEDYNAMICS represents intermediate results, the input and output rules enforced by its operations, the records retained for each data source, the procedures used for validation, the executable examples, and the boundaries among software revisions. It is grounded in the public 0.1.0rc3 artifact and later repository evidence available on September 25, 2026. The public artifact is still a prerelease rather than final 0.1.0, and later work on `main` must not be attributed to rc3 without an exact revision. The supplement distinguishes implemented behavior from older compatibility interfaces, names reserved for future work, experimental source candidates, provider availability recorded on the evidence date, historical rc1 findings, and post-release analyses limited to explicitly stated data and methods. Its purpose is to make the evidence beneath the manuscript’s concise analytical examples directly inspectable.

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S1 Purpose, scope, and version boundary

The main manuscript develops a scientific argument: computational repeatability does not necessarily reveal what an analysis asked of environmental observations. This supplement provides the repository-level detail beneath that argument. It documents the grammar reviewed on September 25, 2026, what each intermediate result represents, the source catalog, validation philosophy, examples limited to stated scopes, and known release limitations without treating the repository as evidence for claims it has not tested.

Four software states must remain distinct:

Table S1: Software states referenced by the manuscript and supplement.

State	What it establishes	What it does not establish
Historical 0.1.0rc1 artifact	The first public prerelease and the outside-user findings that motivated hardening	Inclusion of later semantic, packaging, documentation, or serialization repairs
Public 0.1.0rc3 artifact	An installable package and source archive were published on PyPI on September 2 from source commit <code>f777eb07d3ad</code> ; first-use repairs, reporting of the imported software revision, explicit time-interval checks, and defined event operations are included	Final 0.1.0 status or inclusion of later changes on <code>main</code>
Post-rc3 repository state	Later synchrony analysis limited to stated data and methods, associated diagnostics, and expanded validation evidence	Automatic inclusion in rc3, arbitrary many-dataset synthesis, national scalability, regime classification, or a stable release
Future final release	A possible archived and citable stable release after remaining scholarly and release gates pass	No final release DOI existed at the evidence date

The public rc3 artifact provides the immutable release boundary for published behavior. Repository runtime, tests, generated references, and validation records provide dated evidence for later work, but `main` remains mutable until an exact commit and evidence bundle are archived. Narrative pages and this supplement interpret that evidence but do not supersede it. Where a behavior remains under development or experimental, it is labeled as such rather than inferred from a roadmap or placeholder name (Tuff, 2026a,g,b,c).

S2 Project origin, users, and contribution model

CUBEDYNAMICS developed through a collection of environmental workflows rather than from an attempt to design a universal language in advance. The repository contains climate and vegetation sources, fire objects, terrain, road, and streamflow lessons, synchrony methods, decision vignettes, and source-quality records. Across those projects, the recurring need was a small shared layer above ordinary scientific Python: environmental information should enter through a recognizable scientific name, transformations should appear in analytical order, and the object and evidence produced after each stage should remain inspectable.

The primary intended users are environmental researchers and research software engineers who already work with xarray and related libraries. A secondary audience includes source maintainers, domain-method developers, educators, reviewers, and developers of automated workflows whose tasks and authority are explicitly limited. The package is not intended to replace numerical arrays, storage systems, catalogs, workflow schedulers, or scientific judgment.

Contribution is separated by scope. A project-specific transformation can remain an ordinary Python function and does not require a new data class or plugin registration. A project-specific data name should call a trusted loader and preserve dimensions, units, coordinate reference system, source identity, and records of origin and processing. Promotion into the shared vocabulary requires a stronger package-level record because public names and input and output rules will be reused outside the originating project. The maintained contribution path therefore joins implementation with direct and piped tests, tests of expected rejection, reviewed source data where applicable, generated reference material, and an executable example (Tuff, 2026a,i).

Table S2: User and contributor pathways in the current project architecture.

Role	Immediate use	Contribution pathway
Environmental researcher	Place a readable inspected sequence around existing xarray work	Report scientific contracts and boundaries through a vignette or project verb
Research software engineer	Maintain reusable workflows, tests, source and processing records, and release evidence	Improve core input and output rules, source adapters, quality assurance, documentation, or packaging
Source maintainer	Expose one vetted product through a scientific noun	Supply provider identity, standardized metadata, checked example datasets, quality assurance, and serving history
Domain-method developer	Keep project assumptions in a named operation without replacing the core	Publish ordinary Python functions, direct and piped tests, and interpretation guidance
Reviewer or collaborator	Descend from expression to the current result description, operation history, source, and evidence	Evaluate whether the recorded choices support the stated question

S3 The inspectable grammar as a layered architecture

In programming-language theory, a grammar defines symbols and rules for combining them into well-formed constructions (Aho et al., 2006). CUBEDYNAMICS uses the term operationally for its environmental vocabulary and documented composition rules; it is not a complete formal language. Python defines an expression as syntax that evaluates to a value (Python Software Foundation, 2026). In the line below, the right-hand side is the expression, while the complete line would normally assign its resulting pipe object to a name:

```
pipe(noun) | verb(parameters) | verb(parameters)
```

A *noun* is a named kind of environmental information, and its *source qualifier* states the provider and product. The implementation sometimes stores this qualifier in a field called `source_flavor`; the paper uses *source qualifier* for the concept. Observation and metadata models provide precedent for retaining the observed property, procedure, source, units, and coordinate context rather than treating a shared variable name as measurement equivalence (Madin et al., 2007; Janowicz et al., 2019; Hassell et al., 2017).

A *verb* is a named transformation whose parameters state choices such as dimension, direction, threshold, quantile, or duration. A Python *callable* is an object that can be invoked with arguments; ordinary functions and methods are common callables (Python Software Foundation, 2026). A *semantic contract* documents accepted inputs and the package’s declared interpretation of the output. *Semantic state* records that interpretation and the information believed to remain, while a *trace* records completed operations, parameters, messages, and state changes in order. These records do not verify the truth of

incoming metadata or the scientific suitability of a method. Semantic workflow research provides related precedent for attaching domain meaning and constraints to computational composition (Bowers, 2012).

Observation support is the spatial area or time interval represented by a value, not merely its coordinate label (Gotway and Young, 2002). A *fixture* is a small, fixed example dataset retained for repeatable offline tests. *Bounded evidence* means evidence limited to an explicitly stated source, query, extent, period, software revision, and test. *Canonical* denotes the maintained public representation expected from an operation; it does not mean that every scientific relationship has the same fields. Here *semantic* refers to scientific interpretation and does not claim implementation of a formal ontology. Source records, provenance, and quality-assurance evidence limited to stated queries and versions ground the compact expression (Tuff, 2026a).

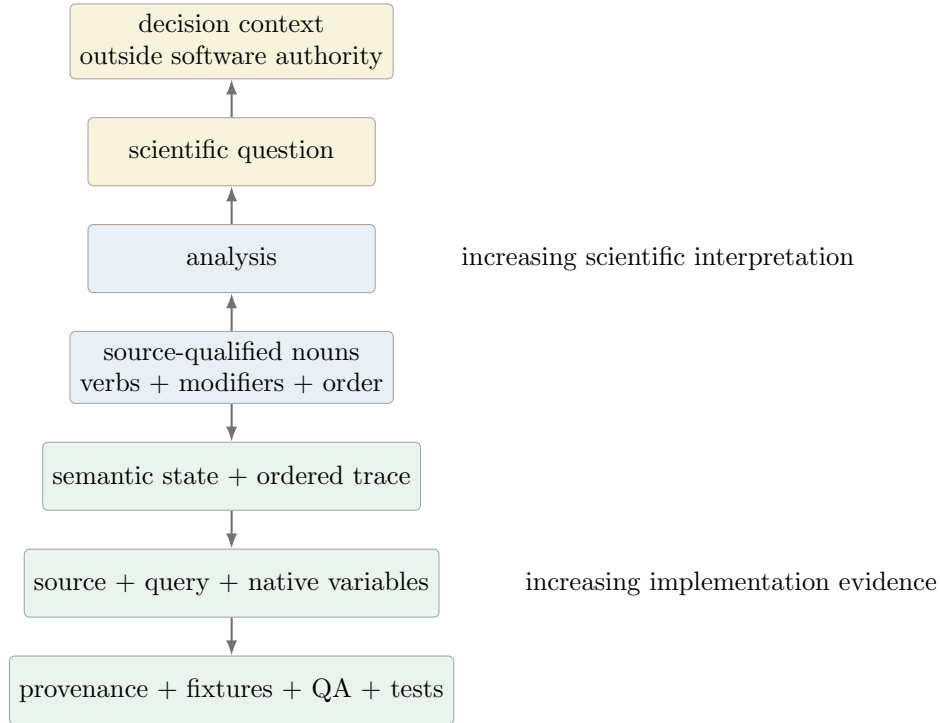


Figure S1: The grammar is an interface between scientific interpretation and supporting evidence. CUBEDYNAMICS exposes analytical choices but does not assume decision authority.

The design maxim follows directly from this architecture: *the shorter the analytical expression becomes, the stronger the evidence underneath it must become*. Concision is useful only if a reader can descend from a noun or verb into its source identity, parameters, state transition, trace, provenance, and validation boundary.

In this paper, *inspectability* is operational rather than perceptual. It means that the software exposes a recoverable record of the source qualifier and query, selected variables and units, explicit parameters, authored operation order, semantic states and transitions, trace messages, and bounded supporting evidence. This definition does not establish that a source or method was scientifically appropriate, that a reader will understand the analysis more quickly, or that the result is suitable for a decision. Those questions require scientific review and, for usability claims, comparative study.

S4 Semantic state and trace

S4.1 State vocabulary

Semantic state is metadata containing the package’s declared or metadata-inferred interpretation of the current scientific object, not a replacement container hierarchy or verification of the object’s scientific truth. Inference reads public metadata and coordinates and does not compute array values solely to infer meaning. Observation ontologies and the CF data model provide related precedents for distinguishing an observed property, result, procedure, units, and coordinate context (Madin et al., 2007; Haller et al., 2019; Hassell et al., 2017). The maintained runtime vocabulary is summarized in Table S3 (Tuff, 2026a).

Table S3: Maintained semantic-state vocabulary.

State	Interpretation
<code>observation</code>	A value whose more specific analytical role has not yet been identified
<code>continuous_field</code>	Numeric measurements distributed in space and/or time
<code>categorical_field</code>	Labels or categories distributed in space and/or time
<code>condition</code>	A true/false scientific state, optionally accompanied by meaningful magnitude and threshold fields
<code>event</code>	One or more identified intervals or occurrences
<code>feature</code>	A discrete geometry or observation feature
<code>relationship</code>	A comparison, association, overlap, or synchrony result
<code>summary</code>	A reduction that no longer retains all source variation
<code>network</code>	Connected features with graph or flow semantics

The state also records dimensions, shape, units, coordinate reference system (CRS), temporal and spatial status, time ordering, remaining temporal variation, source qualifier, and whether source provenance is present. These fields allow a verb contract to ask whether required information remains available. Absent, stale, or contradictory input metadata remain a limitation: contracts can inspect recorded metadata but cannot independently establish that a provider description is correct.

S4.2 Trace boundary

Each completed stage is appended to a trace object during pipe construction with its operation name, parameters, messages, and state transition. The trace covers the inspected statement only. Preparation before `pipe` and work after `unwrap` remain outside it, so the trace complements rather than replaces environment capture, source snapshots, workflow provenance, and revision control (Davidson and Freire, 2008; World Wide Web Consortium, 2013; Soiland-Reyes et al., 2022; Leo et al., 2024).

```
print(analysis.explain())
print(analysis.semantic_state.as_dict())
print([step.as_dict() for step in analysis.semantic_trace])
report = analysis.validate()
result = analysis.unwrap()
```

The inspection methods do not rearrange the expression, add analysis stages, certify source observations, or compute lazy arrays merely to infer semantics. `unwrap` is a local boundary: it returns the wrapped value to ordinary Python without forcing computation or certifying the wider workflow.

Runtime identity is part of the inspection boundary. `version_info()` reports the package version, imported path, artifact kind, and Git revision when version-control metadata are available. This helps

distinguish a public wheel from a later development checkout or a stale notebook kernel; it does not prove that the wider environment or input data have been archived.

S4.3 Branch and custom-callable boundaries

Composition retains less than complete end-to-end lineage. In the Working Lands example, the warm and dry branches each have their own state and trace before they are unwrapped. The final **overlap** stage records the condition labels, selected variables, and exact-alignment requirement, but its result does not embed both complete upstream traces and source records. Reconstructing the full analysis therefore requires retaining the branch records alongside the final pipe.

The **apply** stage is an explicit extension boundary. It forwards the current object to a user callable, whose code controls the output’s values, dimensions, and shape; CUBEDYNAMICS then infers semantic state from the returned public object. Pipe syntax and post-operation inspection remain available, but the grammar cannot infer or validate the callable’s scientific intent unless the project supplies an appropriate contract and metadata. The current trace does not by itself archive the callable’s source code, package version, or revision, so durable reconstruction requires those identifiers in the surrounding project provenance. Thus extension without subclassing is supported, but arbitrary callables do not automatically acquire maintained semantic guarantees.

S5 Authored order and semantic constraints

S5.1 Order categories

Order knowledge explains a sequence but never rewrites it. Semantic workflow systems provide precedent for constraining meaningful workflow composition without treating every executable chain as scientifically valid (Gil et al., 2011; Miksa and Rauber, 2017). The repository distinguishes the categories in Table S4 (Tuff, 2026f,g).

Table S4: Categories used to describe operation-order relationships.

Category	Meaning
REQUIRED_ORDER	A later operation requires the scientific object produced by an earlier operation
ORDER_CHANGES_MEANING	Both orders may execute but answer different questions
ORDER_REMOVES_REQUIRED_INFORMATION	An earlier operation can erase information required later
ORDER_EQUIVALENT_OR_NEAR_EQUIVALENT	Equivalence holds only under stated assumptions

The manuscript’s threshold/mean example is intentionally executable in both orders. Thresholding a field and then averaging asks for the proportion of observations satisfying the condition. Averaging and then thresholding asks whether the reduced value satisfies a newly defined condition. The current grammar preserves both statements and records that their order changes meaning.

Other sequences are rejected because information is absent or has been removed. Event detection requires a temporal condition with ordered time and remaining variation. A retained length-one time coordinate after reduction does not restore the removed variation. The grammar therefore rejects event detection after such a reduction and recommends detecting events earlier.

S5.2 Conditional equivalence

For a fixed climatology C , identical linear spatial weights, and identical masks,

$$\text{mean}_s (T(s, t) - C(s, t)) = \text{mean}_s T(s, t) - \text{mean}_s C(s, t). \quad (1)$$

The equality need not hold if climatologies, masks, interpolation, missingness, or weights differ between stages. The software records the assumptions associated with a conditional rule rather than treating a mathematical special case as universal permission to rearrange a workflow.

S5.3 Event scope is explicit

Event detection begins with a temporal condition that retains ordered variation. The resulting events are local to the indexed cells unless the analyst explicitly consolidates them. The post-rc1 event vocabulary therefore separates `detect_events`, `consolidate_events`, and `event_metrics`. Consolidation requires user-declared temporal-gap and spatial-connectivity rules; metrics retain whether they describe local-cell events or consolidated regional episodes. Seasonal gaps are not silently bridged, and a regional episode is not inferred merely because multiple cells are active (Tuff, 2026g).

S6 Condition, reduction, and metadata contracts

S6.1 Canonical condition representation

The post-RC1 implementation uses one public condition representation across thresholding, quantile classification, and logical overlap. This removes an implementation-specific `DataArray`/`Dataset` branch from ordinary composition while retaining only fields with defined scientific meaning.

Table S5: Canonical condition and summary contracts.

Operation	Public object	Semantic content
<code>threshold.state</code>	Condition Dataset	Boolean state ; may include meaningful magnitude and threshold
<code>quantile.state</code>	Condition Dataset	Boolean state ; empirical threshold and associated magnitude may be retained
<code>overlap</code>	Condition Dataset	Boolean state only; no fabricated magnitude or threshold
Condition $\rightarrow$ mean	Summary Dataset	Reduced floating-point state interpreted as occurrence proportion; auxiliary definition arrays are not averaged
Summary $\rightarrow$ <code>threshold.state</code>	New condition Dataset	A new explicit threshold defines a condition on the aggregate

The key invariant is that semantic metadata must describe the result, not merely survive from the input. A condition mean is therefore a summary rather than a Boolean condition. Its state values represent the proportion of retained observations for which the condition was true. A scientist who wants a reduction of magnitude or threshold arrays must request that question explicitly (Tuff, 2026f,g).

S6.2 Comparator and missing-value semantics

Threshold directions are inclusive: **above** uses greater-than-or-equal and **below** uses less-than-or-equal. The current condition contract is two-valued. When an input value is missing, the numerical comparison evaluates to false in the Boolean **state**, while a retained magnitude remains missing. A subsequent unweighted arithmetic mean of **state** therefore uses all indexed positions along the named dimension as its denominator and counts the missing-input position as a non-occurrence rather than excluding it as unknown. The resulting value must not be described as a proportion among valid observations unless the input is known to be fully finite. The implementation does not currently expose a three-valued condition, validity mask, minimum-valid-observation rule, area or exposure weighting, or a CUBEDYNAMICS-specific quantile interpolation parameter; **quantile_state** delegates the empirical quantile calculation to xarray’s default method. Sample-quantile definitions differ across statistical conventions, so this implicit method choice should remain visible when ties or short samples are consequential (Hyndman and Fan, 1996). The reviewed Working Lands fixture is fully finite, so the missing-value limitation does not affect its reported frequency.

S6.3 Core verb contract matrix

Table S6 summarizes the contracts most directly supporting the manuscript’s claims. It is intentionally narrower than the complete public namespace inventory in Section S13.3.

Table S6: Core semantic contracts and their principal boundaries.

Operation	Required input	Result	Principal semantic or evidentiary boundary
anomaly	Numeric field with named dimension	Continuous field	Uses the mean along the named dimension; the comparison baseline remains an analytical choice
mean	Field, summary, or condition with named dimension	Summary	Uses skip-missing reduction for numeric fields; condition input reduces Boolean state to occurrence proportion
variance	Numeric field with named dimension	Summary	Squares known units; unknown units remain unknown
threshold_state	Numeric field or summary	Condition Dataset	Inclusive comparison; missing input becomes false in the two-valued state
quantile_state	Numeric field with ordered temporal variation	Condition Dataset	Cell-wise empirical threshold through xarray defaults; Dask execution gathers the selected time axis into one chunk while preserving spatial chunks

Operation	Required input	Result	Principal semantic or evidentiary boundary
<code>align_time</code>	Two temporally indexed objects	Aligned value plus temporal report	Makes label and support choices explicit; does not resample or alter scientific values
<code>overlap</code>	Two condition Datasets	Condition Dataset	Requires exact coordinates and checks known temporal support; records composition metadata but not complete upstream branch histories
<code>detect_events</code>	Ordered temporal condition with remaining variation	Event result	Rejects continuous input or input whose temporal variation has been removed
<code>consolidate_events</code>	Local-cell event result plus declared connectivity	Regional-episode result	Temporal gap and spatial connectivity are user choices, not inferred scientific truth
<code>event_metrics</code>	Event result with explicit scope	Event-summary result	Retains whether metrics describe local cells or consolidated regional episodes
<code>plot</code>	Supported semantic object and dimensionality	Rendered view while retaining the value	Ambiguous general Datasets require explicit variable selection; optional backends remain environment dependent
<code>apply</code>	Any object accepted by the user callable	Callable-defined object	Output state is re-inferred, but scientific intent and shape changes are not validated by a maintained verb contract

S6.4 Units after transformations

Transformations update scientific metadata as well as numerical values. Variance deterministically squares known physical units: `degC` becomes `degC^2`, `mm` becomes `mm^2`, and compound units are enclosed before squaring. Dimensionless variance remains dimensionless, while missing or explicitly unknown units are not invented. This contract preserves the source and provenance record while changing the interpretation of the statistic.

S7 Alignment, interoperability, and visualization

S7.1 Alignment is part of the question

Combining scientific operands requires explicit alignment. Logical overlap requires exact coordinate agreement, while harmless differences in dimension ordering can be normalized. Diagnostics distinguish incompatible dimension names, temporal coordinates, and spatial coordinates. The operation does not silently reproject, resample, shift, truncate, or intersect operands because those choices alter the analysis. Exact coordinates are necessary for this operation but are not evidence that the operands share observational footprints, temporal support, calendars, measurement processes, uncertainty, or scientific comparability. This boundary is consistent with the spatial change-of-support literature, which shows that apparently compatible data can still answer different questions when their scale or support differs (Gotway and Young, 2002; Dungan et al., 2002; Baumann, 2024).

The presence of an implemented `align_cube` stage does not weaken this principle. Alignment is a visible analytical operation with a chosen reference, not an implicit side effect of a later combining verb. Computational compatibility after explicit alignment still does not establish measurement equivalence.

S7.2 Temporal labels are not observation support

The current grammar distinguishes a timestamp from the interval represented by the observation. CF metadata provide related conventions for describing coordinates, bounds, and data-model relationships (Hassell et al., 2017). `TemporalSupport` records known support metadata, while `TemporalAlignmentReport` records whether labels and support are compatible. The `align_time` stage supports an explicit label-alignment mode and a stricter mode that requires exact known support. It does not resample values, shift coordinates, or invent support when metadata are absent. A support mismatch therefore remains a scientific choice to resolve, and unknown support remains visible as a review condition rather than being silently treated as agreement (Tuff, 2026a,g).

This distinction matters even for nominally daily products. PRISM and gridMET can assign the same date labels while representing different daily support conventions. Source adapters retain those conventions so coordinate equality alone cannot authorize combination. The Working Lands example avoids this cross-source issue because both branches use the same PRISM daily product.

S7.3 Ordinary scientific Python objects

Nouns return ordinary xarray objects, conditions remain xarray Datasets, and `unwrap` returns the current value. NumPy- and Dask-backed execution behavior remains with the underlying scientific Python stack when an operation permits it. A lazy output does not by itself prove bounded remote access: transport behavior is validated separately from array evaluation.

S7.4 Plotting dispatch

The plotting interface selects a representation from semantic object and dimensionality rather than requiring the scientist to strip metadata or manually extract the common condition field. Current intended dispatch is summarized in Table S7.

Table S7: Semantic plotting dispatch in the post-RC1 repository state.

Input	Default interpretation
Three-dimensional spatiotemporal field	Interactive cube view
Two-dimensional spatial summary	Static map
One-dimensional temporal summary	Line plot
Condition Dataset	Canonical <code>state</code> field
Event Dataset	<code>event_active</code> where appropriate
General ambiguous Dataset	Explicit variable selection required

Lexcube, an optional interactive viewer for multidimensional data cubes, remains a visualization extra. The core architecture therefore places the semantic object before the rendering backend. The custom cube viewer and ordinary static plotting do not require Lexcube (Tuff, 2026f,g).

S8 Source-qualified noun inventory

S8.1 Production-catalog nouns

Table S8 reports the source flavors in the maintained production discovery catalog. A common noun is a common entry point, not a claim that its source flavors are observationally interchangeable (Tuff, 2026a,e).

Table S8: Implemented production-catalog nouns and source flavors.

Noun	Source flavors	Scope note
<code>temperature</code>	<code>prism</code> , <code>gridmet</code>	Source-specific statistics, units, grids, revisions, and provenance retained
<code>precipitation</code>	<code>prism</code> , <code>gridmet</code>	Daily precipitation access under source-specific contracts
<code>vpd</code>	<code>gridmet</code>	Variable-specific scientific QA remains necessary beyond catalog presence
<code>wind</code>	<code>gridmet</code>	Native interpretation and units retained
<code>humidity</code>	<code>gridmet</code>	Native interpretation and units retained
<code>radiation</code>	<code>gridmet</code>	Native interpretation and units retained
<code>surface_reflectance</code>	<code>sentinel2</code>	Spectral, spatial, acquisition, and cloud/scene constraints retained
<code>vegetation_index</code>	<code>sentinel2</code>	Derived index linked to its reflectance inputs and source record

PRISM and gridMET temperature illustrate the source-qualification rule. Table S9 preserves the detailed comparison moved from the main manuscript. The products use different construction methods and native grids (Daly et al., 2008; Abatzoglou, 2013).

Unit conversion or regridding may make these objects computationally compatible without demonstrating scientific equivalence. Observation identity and fitness for purpose still depend on the product, procedure, statistic, support, provenance, and intended analysis (Madin et al., 2007; Janowicz et al., 2019; Whitfield, 2012).

Table S9: Selected distinctions retained beneath the shared **temperature** noun.

Property	PRISM flavor	gridMET flavor
Daily statistics	Maximum, minimum, and mean	Maximum and minimum; no native mean
Native variables and units	tmax , tmin , tmean ; degrees Celsius	tmmx , tmmn ; kelvin
Grid and documented span	Approximately 4 km; 1981–present	1/24°; 1979–present
Daily temporal support	Product-specific support retained in metadata	Different product-specific support retained in metadata
Revision behavior	Recent daily grids may be remodeled	Recent observations progress through maturity stages
Reviewed QA scope	Bounded temperature extract	Bounded maximum-temperature extract

The gridMET names **wind**, **humidity**, and **radiation** are likewise umbrella entry points, not complete scientific definitions. The selected native variable, units, measurement height or aggregation where applicable, query, and source revision must remain in the returned metadata and provenance. Catalog presence alone does not establish that every possible interpretation of the noun is implemented or reviewed.

S8.2 Installed bounded source candidates

The repository also contains three installed candidate projects that exercise different source shapes: streamflow, elevation, and roads. The roads project exposes two bounded providers, so Table S10 contains four source rows. In this section, *area of interest* (AOI) denotes the explicitly requested spatial extent. These projects are documented and vignetted, but remain outside production source discovery and have no production serving revisions (Tuff, 2026d).

Table S10: Installed source candidates outside the production catalog.

Candidate	Implemented bounded scope	Explicit boundary
USGS streamflow	One station, continuous discharge, native status and units, time $\times$ station	Bounded dates, batches, observations, retries, and response bytes; not hydrologic decision validation
USGS 3DEP elevation	One fully covering CONUS 1/3 arc-second tile, native static raster	No mosaicking, silent clipping, value reprojection, or invented time axis
Overture roads	Explicit release with native road features and classes	Bounded AOI, partitions, row groups, features, attempts, and bytes
OpenStreetMap roads	Small Overpass way query with native tags and identifiers	Bounded AOI and response; not a production application backend or complete routable network

These candidates demonstrate that terrain, networks, and station time series can enter the broader grammar without being forced into one raster-cube shape. Their source records should cite the USGS Water Data for the Nation database, 3DEP accuracy assessment, and OpenStreetMap’s source and quality literature; Overture should be tied to the exact tested release and schema (U.S. Geological Survey, 2026; Stoker and Miller, 2022; Haklay and Weber, 2008; Haklay, 2010). They do not justify a universal source abstraction, national certification, cross-source quality ranking, or automatic promotion into the public source catalog.

S9 Source QA and evidence layers

S9.1 Evidence has distinct jobs

Source evidence is separated because different checks answer different questions:

Table S11: Evidence layers and their claim boundaries.

Evidence layer	Supports	Does not establish
Retrieval and live health	The requested endpoint and access behavior were available at a recorded time	Continued availability or scientific correctness
Fixture and provenance	The reviewed bytes, query bounds, upstream identity, and checksum	All products, places, periods, or provider observations
Structural interpretation	Variables, dimensions, coordinates, units, CRS, masks, and schema	Observational accuracy or decision fitness
Numerical QA	Broad ranges, finiteness, ordering, and source-specific relationships	Independent ground truth or causal validity
Visual review	Detectable spatial, temporal, orientation, clipping, or rendering problems in reviewed outputs	Universal scientific validity
Semantic tests	State transitions, operation order, metadata updates, and expected rejection	Completeness of the scientific rule catalog
Publication checks	Executable notebooks, generated-reference freshness, documentation links, and rendering	User productivity or superiority to another workflow

A serving revision identifies a CubeDynamics adapter and interpretation contract, not the provider's scientific version. Schema fingerprints expose the interpreted structure without forcing value computation. Offline fixture certification and live endpoint health remain separate records.

S9.2 Phase 1 production-source evidence

Table S12: Bounded Phase 1 source evidence recorded by the repository.

Source evidence	Serving revision	Reviewed boundary
PRISM temperature	<code>temperature.prism@2026-08-26.1</code>	Checksum-controlled bounded temperature extract; pass with documented caveats
gridMET maximum temperature	<code>temperature.gridmet@2026-08-26.1</code>	One bounded maximum-temperature extract; annual-file retrieval remains a caveat
Sentinel-2 B04/B08 and normalized difference vegetation index (NDVI)	<code>surface_reflectance.sentinel2@2026-08-26.1</code>	One bounded South Dakota window; pixel-level cloud masking is not implemented in the baseline

The PRISM profile checks checksum and source flags, explicit CRS, increasing dates, finite nonempty arrays, broad physical bounds, coordinate overlap with the requested AOI, and the relationship that

minimum temperature does not exceed maximum temperature. The gridMET profile additionally checks its expected $1/24^\circ$ spacing, orientation, and kelvin range. The Sentinel-2 profile checks B04/B08 presence, ordered acquisition dates, finite pixels, 10 m spacing, native UTM coordinates, provider scaling, and NDVI bounds. Sentinel-2 mission characteristics are described by [Drusch et al. \(2012\)](#). These are bounded adapter checks, not certifications of every variable or observation ([Tuff, 2026e](#)).

S9.3 Live-source status

The live-source record accessed September 25, 2026 reports that a bounded PRISM request passed with documented caveats. Daymet is a separate credentialed live-certification target rather than a production-catalog noun or one of the installed candidates in Table S10; it is recorded as **BLOCKED** when the required Earthdata token is absent. Daymet’s data record includes product-specific uncertainty information, but a missing local credential remains an access boundary rather than an observational judgment ([Thornton et al., 2021](#)). Provider availability, authentication, package correctness, provenance retention, fixture validation, live health, and scientific suitability must remain distinguishable ([Tuff, 2026g](#)).

S10 Executable vignette inventory

The repository designates 12 supported offline notebooks. Each supported vignette declares a checksum-controlled observational extract, provenance record, offline execution, and required portable plot output. The collection contains eight core grammar lessons, three source-shape lessons, and the Working Lands worked environmental vignette ([Tuff, 2026i,g](#)).

Table S13: Supported executable vignette inventory.

Notebook	Role	Primary teaching purpose
<code>cube_from_arrays.ipynb</code>	Core lesson	Construct a labeled scientific cube from NumPy data with coordinates, units, and provenance
<code>cube_from_tidy_table.ipynb</code>	Core lesson	Reshape tabular observations into a comparable cube
<code>cube_from_dataset.ipynb</code>	Core lesson	Work with a multi-variable xarray Dataset
<code>grammar_basics.ipynb</code>	Core lesson	Introduce nouns, verbs, pipes, and inspection methods
<code>verbs_gallery.ipynb</code>	Core lesson	Demonstrate implemented transformations and summaries with visible outputs
<code>states_and_events.ipynb</code>	Core lesson	Move from continuous observations to conditions and events
<code>custom_verb_project.ipynb</code>	Core lesson	Use an ordinary project-owned callable without a required base class or registration
<code>lazy_composition.ipynb</code>	Core lesson	Preserve Dask-backed deferred execution where operations permit it
<code>elevation_landscape.ipynb</code>	Source-shape lesson	Analyze a static terrain raster without inventing a time dimension

Notebook	Role	Primary teaching purpose
<code>roads_local_network.ipynb</code>	Source-shape lesson	Retain native road-network source semantics and explicit spatial operations
<code>streamflow_snapshots.ipynb</code>	Source-shape lesson	Replay rolling station observations with native status and snapshot identity
<code>working_lands.ipynb</code>	Worked environmental vignette	Compose PRISM temperature and precipitation conditions into a bounded co-occurrence frequency

The vignette contract is narrative as well as computational: context, question, compact pipe, figure, interpretation, data record, and reproduction instructions must remain visible. Generated or random measurements are not silently substituted when remote services are unavailable.

S11 Working Lands protocol and interpretation

S11.1 Question and observations

The executable Working Lands vignette asks where, within a bounded central South Dakota window during July 2024, maximum temperature was at or above each cell’s empirical 0.75 quantile while daily precipitation was at or below 0.1 mm. Both nouns use PRISM observations. These are illustrative screening definitions rather than externally validated impact thresholds. The AOI keeps the acquisition and notebook bounded and was not selected as a known impact hotspot (Tuff, 2026j).

Table S14: Frozen Working Lands analysis input.

Field	Recorded value
Provider/product	PRISM Climate Group; AN81d/AN91d daily time series
AOI	[−101.2, 43.7, −100.4, 44.3] in WGS84
Dates	July 1–31, 2024, daily
Variables	Daily maximum air temperature (degrees Celsius); daily precipitation total (mm)
Cube size	$31 \times 15 \times 19$ for each noun
Temperature range	21.922–40.386 °C
Precipitation range	0–40.260 mm/day
Finite cells	8,835 for each noun
Fixture	<code>tests/fixtures/real_data/sd_working_lands_july_2024.nc</code>
Fixture SHA-256	<code>f9f3f0da6c621383b60d4895e661a185d3d58e7393b4f339ee91d36d83228a6a</code>
Generated measurements	None

S11.2 Analytical statement

```
warm_branch = (
    pipe(temperature)
    | v.quantile_state(
        quantile=0.75, direction="above", name="warm_july_day"
    )
)
```

```

dry_branch = (
    pipe(precipitation)
    | v.threshold_state(
        threshold=0.1, direction="below", name="trace_or_no_rain"
    )
)

warm = warm_branch.unwrap()
dry = dry_branch.unwrap()

analysis = (
    pipe(warm)
    | v.overlap(dry, name="warm_and_dry")
    | v.mean(dim="time", keep_dim=False)
)
frequency = analysis.unwrap()["state"]
frequency_percent = frequency * 100 # presentation outside the trace

```

The quantile and fixed-threshold stages create two canonical condition Datasets. The displayed code retains `warm_branch` and `dry_branch` so their traces remain available after the values are unwrapped for composition. The inclusive comparators are greater-than-or-equal for warm and less-than-or-equal for dry. Exact coordinate agreement is required before overlap. Overlap returns a condition with Boolean `state`; temporal mean returns a summary whose `state` is the unweighted proportion of the 31 days satisfying both definitions. Multiplication by 100 occurs after `unwrap()` and is therefore a presentation conversion outside the trace. The branch traces must be retained with the final overlap trace to reconstruct the full analytical lineage.

Table S15: Inspection path retained for the Working Lands statement. This table summarizes the documented records rather than substituting for the exact serialized output required in the frozen submission archive.

Object	Recorded interpretation	Required accompanying record
<code>warm_branch</code>	PRISM maximum-temperature condition	Source query, quantile parameters, messages, and transition from continuous field to condition
<code>dry_branch</code>	PRISM precipitation condition	Source query, inclusive threshold parameters, messages, and transition from continuous field to condition
<code>analysis</code>	Unitless temporal occurrence-proportion summary	Exact-alignment rule, overlap definition, named reduction dimension, and final state
<code>frequency</code>	Percentage used for plotting or reporting	Explicit post- <code>unwrap</code> multiplication by 100, recorded by the surrounding notebook rather than the pipe trace

The fully finite fixture produces a grid spanning 6.5% to 22.6% co-occurrence frequency, equivalent to 2–7 of 31 days. “Warm” means at or above the cell’s own threshold computed from this same July sample using `xarray`’s default quantile method; statistical packages can use different sample-quantile definitions (Hyndman and Fan, 1996). In the absence of ties, this rank-based definition selects approximately one quarter of the days at each cell by construction; spatial variation in the final map therefore primarily reflects when the dry condition coincides with those locally ranked warm days. It is not a long-term climatological anomaly. “Dry” means precipitation at or below the stated daily cutoff. The output is not a drought classification, vegetation response, land-sensitivity measure, causal attribution, harm estimate, economic impact, or risk score. Compound-event frameworks address broader dependencies, drivers, and impacts than this illustrative Boolean co-occurrence screen (Leonard et al.,

2014; Zscheischler et al., 2020).

S11.3 Acceptance checks

The fixture and notebook are rejected if source records indicate generated measurements; dimensions, dates, or units differ from the declared contract; cells are nonfinite or outside broad physical bounds; temperature and precipitation coordinates do not align exactly; required source-QA or decision figures are absent; named public operations are unavailable; direct and piped results disagree; or the documentation build and link checks fail. These checks establish the bounded analytical object and its reproducibility, not its fitness for a management decision.

The submission package should include the actual checksum-bound co-occurrence map generated by this fixture. No presentation mock-up or statewide-looking substitute is treated as empirical evidence. That figure is a remaining assembly step until the reviewed artifact and its outputs are copied into the manuscript archive.

S12 Validation philosophy and RC1 triage

S12.1 Contract-compatible execution and informative rejection

Validation of a scientific grammar cannot be reduced to the fraction of arbitrary operation chains that run. Scientific software often lacks a complete test oracle, so invariants, relational checks, expected failures, and metamorphic relationships can provide evidence where one absolute expected output is unavailable (Kanewala and Bieman, 2014; Barr et al., 2015; Segura et al., 2016). Some apparent failures reveal genuine defects; others demonstrate that the grammar correctly refuses an expression that violates a maintained contract. The post-RC1 process therefore classifies findings before changing behavior (Tuff, 2026f).

Table S16: Representative behavioral checks underlying the semantic claims.

Expression or input	Recorded behavior	Interpretation
$\{0, 2, 4, 6\}^{\circ}\text{C} \rightarrow (T \leq 3) \rightarrow$ mean	0.5	Fraction of cells satisfying the condition
$\{0, 2, 4, 6\}^{\circ}\text{C} \rightarrow \text{mean} \rightarrow$ ($T \leq 3$)	TRUE	Condition evaluated on the regional mean
Boolean condition $\rightarrow \text{mean}(\text{time})$	Floating-point state only	Occurrence proportion; condition-definition arrays are not averaged
Variance of a field in degC	Units become degC²	Metadata follows the statistic
Overlap with unequal time or spatial coordinates	Rejected with axis-specific diagnostic	Alignment is explicit rather than silently inferred
Event detection after time variation is removed	Rejected despite a retained length-one coordinate	Coordinate presence is not confused with remaining variation

S12.2 Finding classification

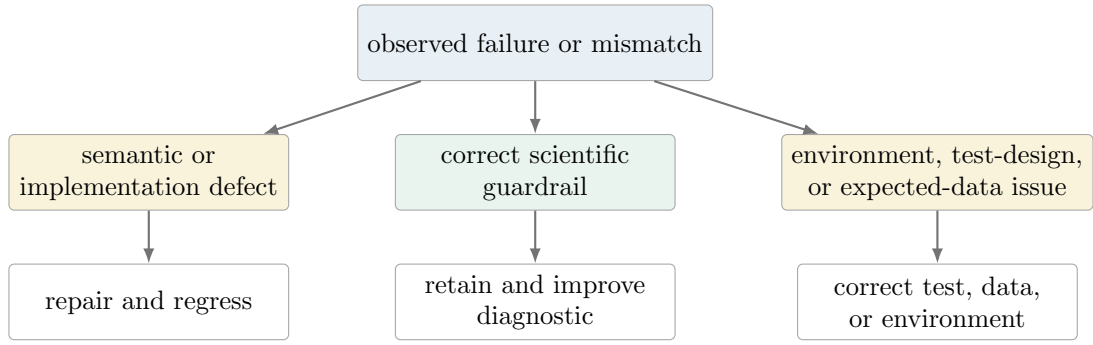


Figure S2: Validation triage distinguishes software defects from correct rejection and from limitations in the test, data, dependency, or environment.

Table S17: Selected RC1 findings and their treatment.

Finding	Classification	Maintained response
Condition mean retained condition semantics and reduced auxiliary definition fields	Semantic defect	Return a summary containing occurrence proportion on state ; retain the condition definition in metadata without averaging auxiliary arrays
Variance retained original units	Semantic defect	Square known physical units; retain dimensionless units and avoid inventing unknown units
Condition and two-dimensional summary plots did not compose	Implementation defect	Dispatch by semantic object and dimensionality
Overlap returned an inconsistent public form	Semantic/interface defect	Return the canonical condition Dataset with Boolean state only
Alignment error did not identify the mismatched axis class	Diagnostic defect	Distinguish dimension-name, temporal-coordinate, and spatial-coordinate mismatches
Mean followed by threshold was rejected although the order lesson taught it	Grammar defect	Support both orders and record that order changes meaning
Exactly misaligned operands were rejected	Correct guardrail	Preserve fail-closed alignment; normalize only harmless dimension ordering
Event detection failed after temporal variation was removed	Correct guardrail	Preserve rejection even if a length-one coordinate remains
Randomly generated arbitrary verb chains failed	Test-design problem	Generate contract-compatible expressions or test expected rejection; do not make the grammar indiscriminately permissive
Fire environmental data did not overlap the event dates	Expected data mismatch	Require temporal overlap; do not substitute or extrapolate climate values

Finding	Classification	Maintained response
Missing optional Lexcube dependency	Optional-environment issue	Keep the integration optional and report the missing extra clearly

S12.3 Dated validation layers

Table S18: Recorded release and development-checkout validation as accessed September 25, 2026.

Check group	Recorded result
Public rc3 artifact	Wheel and source distribution published on PyPI through Trusted Publishing; source attestation records commit <code>f777eb07d3ad</code> (abbreviated)
Unarchived development checkout on <code>main</code> , accessed September 25, 2026	Complete deterministic offline suite: 932 passed, 5 skipped
Focused synchrony/public-API checks	26 passed
Generated-reference, documentation-link, and guide checks	92 passed
Supported real-data vignettes	12 retained as the supported offline notebook suite with required plots
Generated references	84 pages checked against the dated development checkout
Visual evidence	8 results checked; all 28 pages of the bounded Phase 1.5 report visually inspected
Documentation and repository gates	Strict MkDocs, built-site links, schema and compile checks, tracked-size policy, and diff checks passed

These results overlap and should not be summed into a single quality score. The rc3 artifact and later `main` checks also have different revision boundaries. They support specific behavioral, documentation, and publication claims; they do not establish usability, human comprehension, error prevention, scientific superiority, live-provider availability, or decision quality (Tuff, 2026c,g).

The aggregate record reports five skipped tests but the manuscript archive does not yet enumerate their names and reasons. A frozen submission evidence bundle should include the complete machine-readable test report so that readers can distinguish intentional dependency or platform skips from unexecuted core behavior.

S13 Secondary extension record outside the primary paper evidence

The material in this section records examples of later or domain-specific package growth. It is not required to establish the main paper’s claims about authored order, source-qualified nouns, condition composition, the Working Lands example, or rc3 release behavior. For journal submission, the complete callable inventory is better preserved with the versioned generated API documentation; the concise record is retained here so that post-rc3 and project-specific capabilities are not silently attributed to the core demonstration.

S13.1 Bounded post-rc3 synchrony evidence

Later repository work reused an observed-PRISM 20×20 checkpoint to interpret spatial synchrony and panel-change diagnostics without recomputing a national analysis. Event synchronization, event coincidence analysis, and functional climate-network methods provide related but distinct methodological lineages (Quian Quiroga et al., 2002; Donges et al., 2016; Wolf et al., 2020); individual CubeDynamics callables should cite only the method they actually implement. The bounded work added nested-radius summaries and descriptive distance, direction, modality, and geographic-coherence diagnostics. The recorded conclusion is deliberately narrow: the high-IQR stack was broad and directionally organized but remained bandwidth-stable unimodal, not a validated climate regime or transition. It recommends a changed, bounded 100×100 benchmark with retained multidimensional diagnostics; it does not establish national scalability, an optimal scientific radius, or regime classification. This post-rc3 analysis is evidence of an extension path, not part of the Working Lands demonstration or the rc3 release claim (Tuff, 2026g).

S13.2 Fire VASE as a bounded domain extension

Fire VASE is the project’s name for a fire-event visualization and extraction object. It illustrates how a project-specific scientific object can participate in the architecture without redefining the minimal grammar; it is not presented here as a standardized fire-science term. The preferred object route is:

`FireEventDaily → FireHull → attach_environment → plot`

The documented cube-first `v.fire_plot` helper remains a high-level convenience. Older `fired.daily` and `event_id` routes are compatibility interfaces rather than the primary teaching model. Environmental attachment can use named temperature, precipitation, vapor-pressure-deficit, wind, humidity, and radiation fields when they are present.

Temporal overlap is required between the fire object and environmental cube. The implementation does not fill a non-overlapping period with substitute climate data, extrapolate beyond observed dates, or reinterpret an absent variable. Fire VASE is therefore evidence that the grammar can support a specialized domain object; it is not evidence for arbitrary cross-domain composition or automated fire interpretation (Tuff, 2026a,f).

S13.3 Public callable status inventory

The `cubedynamics.verbs` namespace contains grammar stages, direct helpers, compatibility names, and reserved placeholders. Sharing a namespace does not make every callable a configured pipe-stage factory. The following tables record the manuscript-relevant callable inventory, including the temporal and event additions made after rc1. The generated reference pages remain authoritative for the complete current namespace and exact signatures (Tuff, 2026h).

The presence of a placeholder such as `correlation_cube` or `fit_model` is not evidence that the advertised analysis exists. Compatibility names remain available but are not necessarily preferred for new workflows. Project-specific callables can operate in a pipe without registration, subject to the `apply` boundary described in Section S3.3; promotion into maintained shared vocabulary requires separate scientific and software review.

Table S19: Core transformations, conditions, and event callables.

Name	Callable type	Status	Primary role
<code>align_cube</code>	Grammar stage	Implemented	Align a field to an explicit reference cube
<code>align_time</code>	Grammar stage	Implemented	Align time labels explicitly and report observation-support compatibility
<code>anomaly</code>	Grammar stage	Implemented	Express departures from a mean
<code>apply</code>	Grammar stage	Implemented	Apply a user callable while retaining pipe syntax
<code>binary_state</code>	Grammar stage	Implemented	Normalize a binary mask to the condition contract
<code>change_state</code>	Grammar stage	Implemented	Define a condition from lagged change
<code>detect_events</code>	Grammar stage	Implemented	Group consecutive true periods into events
<code>consolidate_events</code>	Grammar stage	Implemented	Consolidate local-cell events using declared temporal and spatial connectivity
<code>event_metrics</code>	Grammar stage	Implemented	Summarize events while retaining local or regional scope
<code>flatten_cube</code>	Grammar stage	Implemented	Stack non-time dimensions into samples
<code>flatten_space</code>	Grammar stage	Implemented	Stack spatial dimensions into pixels
<code>mean</code>	Grammar stage	Implemented	Average over a named dimension
<code>month_filter</code>	Grammar stage	Implemented	Retain selected calendar months
<code>ndvi_from_s2</code>	Grammar stage	Implemented	Derive NDVI from Sentinel-2 red and near infrared
<code>quantile_state</code>	Grammar stage	Implemented	Define a condition from an empirical quantile
<code>threshold_state</code>	Grammar stage	Implemented	Define a named true/false condition
<code>variance</code>	Grammar stage	Implemented	Measure variation with squared-unit propagation
<code>zscore</code>	Grammar stage	Implemented	Standardize relative to mean and spread

Table S20: Comparison, synchrony, and spatial callables.

Name	Callable type	Status	Primary role
<code>block_signature</code>	Grammar stage	Implemented	Summarize a named spatial block through time
<code>collect_blocks</code>	Grammar stage	Implemented	Collect compatible block signatures
<code>compare_blocks</code>	Grammar stage	Implemented	Compare collected spatial-block signatures
<code>duration_synchrony</code>	Grammar stage	Implemented	Compare event duration
<code>occurrence_synchrony</code>	Grammar stage	Implemented	Compare co-occurrence among conditions
<code>overlap</code>	Grammar stage	Implemented	Identify coincident truth in exactly aligned conditions
<code>rasterize_observations</code>	Direct helper	Implemented	Place feature observations on a reference grid
<code>rolling_median_split_synchrony</code>	Grammar stage	Implemented	Measure rolling median-split synchrony
<code>rolling_tail_dep_vs_center</code>	Grammar stage	Implemented	Compare rolling tail behavior with the center
<code>severity_synchrony</code>	Grammar stage	Implemented	Compare magnitude where conditions co-occur
<code>sync_with</code>	Grammar stage	Implemented	Compare with an aligned temporal cube
<code>timing_synchrony</code>	Grammar stage	Implemented	Compare event timing

Table S21: Visualization, Landsat, and output callables.

Name	Callable type	Status	Primary role
<code>climate_hist</code>	Grammar stage	Implemented	Render a climate distribution
<code>diagnostic_panel</code>	Grammar stage	Implemented	Render a diagnostic panel
<code>landsat8_mpc</code>	Grammar stage	Implemented integration	Load Landsat observations through the Microsoft Planetary Computer (MPC); this namespace entry is not a production-catalog noun
<code>landsat_ndvi_plot</code>	Direct visualization helper	Implemented	Render a Landsat NDVI view
<code>landsat_vis_ndvi</code>	Direct helper	Implemented	Prepare Landsat NDVI for visualization
<code>plot</code>	Grammar stage	Implemented	Render according to semantic object and dimension
<code>plot_mean</code>	Grammar stage	Implemented	Render a spatial-mean time series
<code>show_cube_lexcube</code>	Grammar stage	Optional extra	Render with Lexcube while retaining the value
<code>to_netcdf</code>	Grammar stage	Implemented and release-tested	Write condition states through a flagged <code>int8</code> copy while leaving the in-memory semantic object unchanged

Table S22: Fire VASE and geometric domain callables.

Name	Callable type	Status	Primary role
<code>extract</code>	Grammar stage	Implemented	Attach fire-hull climate summaries
<code>fire_derivative</code>	Direct visualization helper	Implemented	Derive a configured fire-event product
<code>fire_panel</code>	Direct visualization helper	Implemented	Render a compact fire-event panel
<code>fire_plot</code>	Direct visualization helper	Implemented	Render one fire event and climate context
<code>fire_vase_panel</code>	Grammar stage	Implemented	Render a multi-event VASE panel
<code>tubes</code>	Grammar stage	Implemented	Construct a tube representation
<code>vase</code>	Grammar stage	Implemented	Construct a configured fire VASE representation
<code>vase_extract</code>	Direct helper	Implemented	Extract cube values for VASE geometry
<code>vase_mask</code>	Direct helper	Implemented	Mask a cube with VASE geometry

Table S23: Compatibility and reserved public names.

Name	Callable type	Status	Primary role
<code>aoi_signature</code>	Grammar stage	Compatibility	Older AOI-signature workflow
<code>compare_aoi_signature</code>	Grammar stage	Compatibility	Older AOI-signature comparison
<code>exceedance</code>	Grammar stage	Compatibility	Alias for <code>threshold_state</code>
<code>vase_demo</code>	Grammar stage	Compatibility	Legacy synthetic demonstration
<code>correlation_cube</code>	Reserved API	Placeholder	Future correlation-cube operation
<code>fit_model</code>	Reserved API	Placeholder	Future modeling operation

S14 Reproduction and evidence commands

The public release boundary can be installed without a source checkout:

```
python -m pip install cubedynamics==0.1.0rc3
python -c "import cubedynamics as cd; print(cd.version_info())"
```

The commands below summarize the repository’s documented reproduction routes for later development evidence. They assume an exact source revision, Python 3.9 or later, and the relevant optional extras installed. The repository declares the MIT License, but this supplement does not provide a frozen operating-system image or resolved submission dependency lock; the commands are therefore not substitutes for an archived environment bundle.

```
# Deterministic offline tests
python -m pytest -m "not integration and not online" -q

# Publication validation and all supported vignettes
python scripts/run_validation.py --run-vignettes
python scripts/run_vignettes.py

# Deterministic decision evidence
python scripts/run_decision_qa.py

# Generated references and documentation
python scripts/build_reference_docs.py --check
```

```
python -m mkdocs build --strict
python scripts/check_site_links.py site
python scripts/check_repository_size.py --mode tracked
```

The validation runner writes machine-readable JSON, PNG evidence, a suite manifest, and a collated PDF under the ignored `artifacts/validation/` directory. The source-QA route is invoked separately with `python scripts/run_source_qa.py`; any credentialed or online result must be labeled with its date and environment rather than grouped with deterministic fixture evidence. Source QA writes to `artifacts/source_qa/`. Deterministic fixture tests cannot establish present endpoint health.

Table S24: Repository evidence map for manuscript claims.

Repository record	Primary use in the manuscript and supplement
PROMPT_LOG.md	Dated implementation decisions, exact validation results, and remaining caveats
docs/project/rc1_validation_triage.md	One-category-per-finding RC1 classification and maintained semantic decisions
docs/concepts/semantic_grammar.md	State vocabulary, verb contracts, order categories, inspection methods, and boundaries
docs/data/phase1_qa.md	Serving revisions, source profiles, bounded source QA, and remaining source limitations
docs/validation/	Publication validation modules, machine-readable evidence, and visual controls
docs/decision_vignettes/working_lands.ipynb	Executable two-noun analysis, code, plots, interpretation, and reproduction instructions
docs/decision_vignettes/validation.md	Working Lands input values, checksum, acceptance criteria, and dependency ledger
docs/reference/verbs/	Generated public callable status and signatures
tests/fixtures/real_data/	Small checked observational fixtures and provenance records

S15 Public rc3 and the resolved rc1 first-use findings

CubeDynamics 0.1.0rc1 was publicly released on August 31, 2026. Its outside-user acceptance exercise reached a **FIX BEFORE HUMAN OUTSIDE TEST** verdict after three first-use findings: core import failed in the tested Debian/aarch64 Python 3.11 image because the eager optional geospatial stack required `libexpat.so.1`; the initially encountered PRISM path implied an unsupported monthly request; and ordinary NetCDF serialization failed on Boolean metadata attributes (Tuff, 2026b; CubeDynamics naive-user testing project, 2026).

Those findings were addressed before the current public candidate. Core grammar and climate-noun imports no longer require the optional Sentinel-2/Rasterio compiled stack; in the exact affected arm64 image, project-run regression evidence records that core import and one bounded PRISM workflow pass while the optional Sentinel-2 path remains unsupported. PRISM nouns default to daily observations, maintained examples state frequency D explicitly, and unsupported monthly access fails rather than silently changing temporal meaning. Raw noun outputs and continuous summaries carry NetCDF-safe scalar metadata. For condition/state Datasets, `to_netcdf` encodes Boolean variables as flagged `int8` in a write-only copy and leaves the in-memory semantic object unchanged (Tuff, 2026g). The available record is not a repeated independent outside-user acceptance exercise for rc3; it should therefore be interpreted as internal release-regression evidence until an independent first-use rerun is archived.

The latest public artifact is 0.1.0rc3, uploaded to PyPI on September 2, 2026 as a wheel and source distribution through Trusted Publishing. PyPI records source attestation commit `f777eb07d3ad` (abbreviated; full identifier in the software citation). Rc3 remains an alpha prerelease rather than final 0.1.0, and later work on `main` is not part of that artifact unless separately archived (Tuff, 2026c).

S16 Claim-to-evidence matrix

Table S25: Claim-level evidence and explicit boundary, part 1.

Claim	Supporting evidence	Boundary
Authored order is preserved	Pipe semantics and order-rule tests; threshold/mean regressions	Does not prove every possible order relationship is cataloged
Both threshold/mean orders are meaningful and executable	Direct and piped examples plus <code>ORDER_CHANGES_MEANING</code> checks	Does not make the results equivalent
Condition objects compose canonically	Threshold, quantile, overlap, reduction, and plotting tests	Does not imply every relationship has magnitude or threshold fields
Condition mean represents an occurrence proportion	State-only reduction and direct-calculation agreement	Unweighted over the named dimension; missing numeric comparisons are false and remain in the denominator
Variance metadata follows the statistic	Unit-propagation regression tests	Unknown units remain unknown
Contract-inadmissible event transitions are rejected	Expected-failure tests after continuous input or removed time variation	Rule catalog is not complete
Local and regional event scopes remain distinct	Event detection, declared consolidation, and scope-retaining metric tests	Connectivity choices remain researcher-specified and do not establish ecological event identity

Table S26: Claim-level evidence and explicit boundary, part 2.

Claim	Supporting evidence	Boundary
Overlap does not silently harmonize operands	Exact-alignment, temporal-support, and axis-specific diagnostic tests	Coordinate or label equality does not establish scientific comparability; scientists must justify alignment explicitly
Nouns retain source identity	Adapter metadata, provenance records, fixtures, and QA profiles	Common noun does not establish source equivalence
Working Lands composes two nouns	Executable notebook, immutable reviewed commit, fixture checksum, direct agreement, and vignette QA	Full lineage also requires separately retained branch records; the checksum-bound map and serialized inspection output are not yet in the submission archive
Visualization follows semantic object and dimension	Plot-selection and rendering regressions	Optional backend availability remains environment dependent
Condition NetCDF export preserves in-memory semantics	Serialization regressions and rc3 first-use gate	Boolean state is encoded in the written copy; export does not certify downstream interpretation
Twelve supported vignettes execute offline	Vignette runner and strict documentation build	Does not certify live providers or all public callables
Dated development validation passed	Recorded focused, offline, documentation, visual, and repository gates as accessed September 25, 2026	Record is mutable until an exact commit and evidence bundle are archived; public rc3 does not contain all later work

S17 Limitations and priorities for a mature supplement

This first supplement is intentionally broad but not exhaustive. The current architecture is primarily linear and demonstrates bounded composition rather than a complete branch-and-join language. A final composed condition does not contain complete upstream histories from separately unwrapped branches. Semantic states simplify scientific meaning and rely on declared or inferred metadata rather than independently verifying them. The present two-valued condition model treats missing numeric comparisons as false rather than representing an explicit unknown state. Exact coordinates do not establish common support or measurement equivalence; temporal-support metadata improve inspection but cannot decide how differing products should be reconciled. The grammar does not represent or propagate product bias, retrieval error, calibration uncertainty, ensemble uncertainty, or covariance. An arbitrary `apply` callable controls its own transformation, so post-operation inference cannot validate its scientific intent or archive the callable’s implementation. Event consolidation requires researcher-chosen connectivity, and the later synchrony diagnostics do not establish regimes or national scalability. Traces do not capture transformations outside the pipe, and source QA ages as providers and products change. Production-catalog nouns do not become interchangeable through a shared name, and candidate adapters are not promoted by documentation alone.

A mature submission supplement should freeze the exact repository commit used for every claim beyond public rc3, archive the machine-readable validation outputs, enumerate skipped tests, and assign persistent identifiers. It should include the checksum-bound Working Lands map, a complete fixture and vignette manifest, rc3 first-use and serialization evidence, final source-certification timestamps, the journal-specific software and data availability form, and any figures or tables moved from the main text during peer review. If the journal permits file bundles, the highest-value addition would be a checksum manifest binding this PDF to the exact fixture provenance, test reports, generated references, figures, public artifact, and later source revision.

The following items are deliberately not claimed here:

- universal environmental-data harmonization;
- arbitrary many-dataset or branch-and-join composition;
- automatic scientific judgment, source selection, or decision suitability;
- observational equivalence among source flavors;
- improved productivity, fewer scientific errors, or better decisions;
- complete workflow provenance from the semantic trace;
- complete branch lineage embedded in the final composed object;
- missing-aware three-valued condition semantics;
- scientific validation of arbitrary project callables;
- live availability inferred from fixture validation;
- final 0.1.0 release status, release DOI, or equivalence between public rc3 and later mutable `main`.

The contribution documented by this supplement is narrower: the repository implements and tests an inspectable connection among source-qualified environmental nouns, semantic transformations, authored order, evolving state, trace, ordinary scientific Python objects, and bounded evidence.

References

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